



ANKARA YILDIRIM BEYAZIT ÜNİVERSİTY
CENG402 MIDTERM REPORT

Group Members (Names and ID numbers)	İrem Şahar 21050141024 Abdullah Yusuf Erdem 22050111077 Cemalettin Seyhan 22050111011
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Supervisor Name	Dr. Öğr. Üyesi Fatma Yerlikaya Özkurt
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1. SUMMARY OF THE PROJECT SCOPE, OBJECTIVES, AND OVERALL PROGRESS

The aim of our project is to develop and implement an early warning and decision-support system that predicts high-mortality diseases in Türkiye using non-laboratory indicators, while also designing an optimization model to support effective patient distribution and efficient hospital resource management. Based on statistics from the WHO, our study focuses on three major diseases that have a significant impact on public health: diabetes, hypertension, and stroke. Our system combines three core components: a unified open-source dataset, a two-stage machine learning pipeline for disease detection and risk assessment, and an optimization model for simulating hospital resource management. All models are trained on publicly available data and rely exclusively on non-laboratory indicators such as lifestyle factors, demographic characteristics, and self-reported symptoms, ensuring the system remains accessible in settings where clinical testing may be limited. The system operates through a two-stage pipeline. In the first stage, a multi-class classification model identifies the patient's dominant pathology. In the second stage, disease-specific risk models generate personalized risk scores, communicated through an intuitive three-tier system: Green (no significant risk), Yellow (behavioral risk patterns requiring attention), and Red (critical probability requiring immediate intervention). Threshold calibration reflects the asymmetric cost of misclassification across diseases, with stroke detection prioritizing recall to minimize missed cases. To test the model using real-world data, a patient survey is currently being conducted. It is being carried out with the approval of the Ethics Committee and in full compliance with all regulations. The sample size was determined in advance through a power analysis and taking into account the prevalence of diseases in Turkey, to ensure that the results are reliable and reflect the actual situation. In addition, we developed and improved an ML pipeline for assessing the risk of diseases (stroke, diabetes, hypertension) based on synthetic medical data by creating more realistic data generation methods that account for outliers. We also improved normalization using Min-Max scaling and implemented a two-level model architecture. Additionally, we introduced a system of dynamic thresholds for patient routing and conducted comprehensive testing (edge, close, and adversarial cases), which increased the system's reliability and practical applicability. We are currently actively working on phase 3 tasks to implement a dynamic hospital routing algorithm that optimizes patient dispatch

2. CLEAR DEFINITION OF THE PROBLEM BEING ADDRESSED AND THE SPECIFIC OBJECTIVES OF THE PROJECT

2.1 Problem of project

Noncommunicable but common diseases such as diabetes, hypertension, and stroke are among the leading causes of death worldwide and place a heavy burden on health systems, particularly in countries with limited resources. Despite significant advances in the use of AI for disease prediction, challenges remain. Firstly, patients often seek help only when symptoms are severe, which worsens the prognosis and increases mortality. Secondly, most existing prediction systems rely heavily on clinical and laboratory data, which are costly, limiting the potential for early screening and prevention. Third problem is that in practice, patients often go to hospitals too late or, conversely, take up laboratory and medical staff resources without real need, leading to hospital overload and inefficient use of medical resources. As a result, healthcare systems face increased workload, reduced quality of service, and professional burnout among medical staff. Our research addresses key healthcare challenges by combining early disease prediction based on machine learning with an algorithm for optimizing resource allocation.

2.2 Specific objectives of project

1. Prepare a structured and clean dataset by merging open-access health datasets focusing on non-laboratory variables such as demographic attributes, lifestyle patterns, and general health measurements. Identify the most influential and sensitive features for each of the three diseases and train a classifier that determines which disease a person is most likely to have.

2. Identify which variables contribute most to risk estimation for diabetes, hypertension, and stroke using feature-importance techniques, including Random Forest importance, features sensitivity analysis.
3. Train and evaluate multiple machine-learning models, including Logistic Regression, Random Forest, and XGBoost and select final models based on accuracy, calibration, robustness, and suitability for tabular health data.
4. Develop optimization modeling that distributes high-risk individuals to nearby medical facilities in conditions of limited capacity. The optimization component will include concepts such as distance-based routing, dynamic capacity changes, and equity-based allocation criteria.
5. Build a deployable web application that integrates classification, risk scoring, and optimization outputs, enabling users to receive personalized assessments through an accessible and structured interface.
6. Measure system performance in various scenarios to evaluate model accuracy, distribution balance and decision support effects.

3. COMPREHENSIVE REVIEW OF RELEVANT EXISTING WORK

The development of machine learning has opened up new possibilities for medicine, ranging from accurate diagnosis to patient monitoring. By analyzing vast amounts of data, algorithms help doctors detect diseases in their early stages and select effective treatments. However, despite the obvious benefits, ML technologies are still rarely used in actual healthcare management. The main barrier is that current research focuses solely on the accuracy of models in laboratory settings, ignoring issues related to their practical implementation in everyday medical practice.

Early work in diabetes prediction has focused predominantly on clinical and laboratory datasets particularly the well-known PIMA Indian Diabetes dataset. For example, **Mujumdar and Vaidehi (2020)** used algorithms such as support vector machines (SVM), decision trees, random forests, and logistic regression. They achieved high accuracy using clinical data such as glucose levels and age. These studies have demonstrated that classical ML algorithms are capable of achieving high accuracy, but remain sensitive to class imbalance. Other studies have applied and describes statistical feature-selection methods for example **Dritsas and Trigka (2022)** developed an approach that used statistical methods such as Pearson's correlation and LASSO regression to identify the most important signs of diabetes: polyuria, polydipsia, and sudden weight loss. We applied these findings in our work by conducting our own analysis of feature importance for diabetes predicting model using the built-in importance mechanisms in XGBoost, which not only improved the quality of the model but also increased its interpretability. Although our diabetes study aligns closely with these works, we took into account the limitations of laboratory models when designing our system, creating a model based on easily obtainable characteristics, which makes it applicable for home or mass screening. First, while many diabetes prediction studies rely on laboratory measurements, our model uses the BRFSS2015 survey dataset, which is available on Kaggle and contains behavioral, demographic, and self-rated health information, allowing risk assessment without the need for clinical tests. This fills an important gap by making prediction systems more accessible to larger populations. Second, after reading various articles on the use of machine learning to predict different models, we concluded that many studies use XGBoost because it handles nonlinear relationships between parameters and unbalanced data well, and this model can be easily configured to prevent underfitting or overfitting. As a result, we built a predictive system based on available characteristics—age, BMI, glucose level, marital status, habits, type of work, physical activity, and other non-invasive parameters—and used XGBoost as a machine learning model, as did **Kanegae et al. (2021)** and **Zhang et al. (2024)**, who demonstrated the effectiveness of XGBoost when using simple physiological and behavioral data, confirming that highly accurate predictions are possible even without laboratory indicators. Third, the integration of probability calibration, threshold optimization using precision-recall curves, and risk stratification

(low/medium/high) is consistent with recommendations in the literature for creating clinically interpretable and practical real-world prediction systems. Furthermore, a recurring issue in medical machine learning literature is the prevalence of class imbalance in real-world registries, where healthy controls vastly outnumber positive cases, often forcing researchers to rely heavily on synthetic oversampling techniques like SMOTE to prevent models from biasing toward the majority class (**Chawla et al., 2002; Kivrak et al., 2024**). Building on these findings, our project adopts the algorithmic strengths of recent literature while fundamentally shifting the data paradigm to address accessibility limitations. In the realm of tabular medical data, research consistently highlights that ensemble learning techniques, particularly XGBoost, outperform single classifiers like Decision Trees in capturing non-linear relationships between risk factors (**Chen & Guestrin, 2016; Yıldız & Kalaycı, 2024**). The traditional approaches for hypertension prediction rely heavily on invasive metrics such as systolic blood pressure readings, total cholesterol, and fasting blood glucose to calculate cardiovascular risk. For example, numerous studies utilizing National Health and Nutrition Examination Survey data have applied algorithms such as Logistic Regression, Support Vector Machines SVM, and Artificial Neural Networks ANN to achieve high diagnostic precision (**Huang et al., 2014; López-Martínez et al., 2020**). However, while these clinically-grounded models achieve high accuracy due to the direct biological correlation of markers like blood pressure with disease capability, they create a significant barrier to entry for early detection, but they require the user to interact with healthcare infrastructure before receiving a risk assessment, rendering them unsuitable for mass home screening or resource-limited settings. So, unlike previous works that necessitate blood panels or professional blood pressure measurements, our model is designed to operate strictly on self-identifiable lifestyle contexts—specifically Salt Intake, Stress Levels, and Sleep Duration—derived from Miadul's synthetic dataset (**Miadul, n.d.**). This approach directly addresses the limitations of laboratory-based models by bridging the gap between daily habits and medical diagnostics. Additionally, contrasting with studies plagued by data imbalance, our selection of a statistically balanced dataset, where 52% positive case distribution allowed us to train unbiased models. By integrating this accessible feature set with precision-recall optimization and a three-tier risk stratification system, our work adheres to modern recommendations for creating explainable, actionable screening tools that empower users prior to clinical intervention (**Sadeghi et al., 2024; Saito & Rehmsmeier, 2015**). Existing literature related to stroke shows that stroke prediction relies on a combination of demographic, metabolic, and cardiovascular variables. Also after researches we saw that well-established medical reports confirm that features such as age, blood glucose level, hypertension status, and cardiac history substantially influence stroke risk (**National Heart, Lung, and Blood Institute, n.d.; Bersano & Gatti, 2023; Nag et al., 2024**). But because these variables interact in nonlinear ways, many researchers note that classical linear models often fail to capture the complexity of stroke-related physiological mechanisms (**Fascia, 2024; Mavrogiorgou et al., 2022**). In our EDA, we revealed distribution plots that older age is strongly associated with higher stroke risk. Elevated glucose levels also increase the risk, especially when combined with hypertension and heart disease, which appear to contribute indirectly. These observations of ours support the conclusion that the major features highlighted in previous research also significantly affect stroke risk in our dataset. Our EDA for chosen stroke dataset further showed that the dataset is highly imbalanced, which complicates the model-training process and makes the usage of classical linear models can't be used.

Optimizing the distribution of patients among hospitals is one of the top priorities in today's healthcare system, especially in the face of high patient volumes, limited resources, and unpredictable external factors. Traditional routing methods in emergency medical services focus primarily on travel distance, without simultaneously accounting for dynamic hospital workload, equipment availability, and external conditions such as weather events or holidays. While this simplicity facilitates implementation, it limits effectiveness in real-world emergency scenarios and fails to ensure an even distribution of the workload. However, in their research,

Zhi-Ying Zhan et al. (2020) demonstrated the delayed impact of hourly changes in precipitation and temperature on ambulance response times, confirming the importance of external factors in emergency situations. In turn, current research trends involve the use of forecasting algorithms and machine learning methods to assess hospital workload and make decisions regarding patient distribution. **Alenany and Ait El Cadi (2020)**, in turn, developed a simulation model to reduce emergency department dwell time and ensured early patient triage based on ML decision trees that predict patients' hospitalization status. However, these approaches require large amounts of historical data and significant computational resources, and they often lack interpretability and are not always applicable in emergency response settings.

Existing studies on sample size estimation for machine learning models actively draw on classical statistical approaches and current epidemiological data. According to the works of **S. Charan and T. Biswas**, a 95% confidence level is considered reliable and generally accepted in biomedical research, while **E. P. Kahraman-Kılbaş and F. Cevahir** indicate that a 5% margin of error is acceptable. As an alternative to classical formulas for sample size planning, the specialized software GPower is successfully used, allowing for accurate power analysis, which is critically important for rare diseases such as stroke that require significantly larger data sets to construct a correct model.

In conclusion, we would like to say that current research aimed at improving healthcare suffers from serious limitations, primarily due to the dependence of models on expensive laboratory data and the fragmentation of research, which separates disease prediction from hospital resource optimization. In addition, many systems unfortunately ignore lifestyle factors, face ethical transparency issues, and lack accessible web interfaces for practical use. Our project addresses these critical gaps by developing an integrated early warning and decision support system based on machine learning. By focusing on easily measurable indicators at home and modeling the allocation of healthcare resources in real time, the proposed study aims to bridge the gap between patient-level diagnosis and system-level management. Thus, in accordance with the analyzed literature, our project also expands the capabilities of classical models by integrating random factors, such as the temporary unavailability of CT scanners and the probability of a vascular center being available. This significantly increases the realism of the simulations and allows for more flexible and adaptive decisions in patient allocation. By connecting patient-level diagnostics with effective system-level management, we ensure not only early disease diagnosis, but also the efficiency and sustainability of hospital systems, especially in resource-constrained settings.

4. DETAILED DESCRIPTION OF THE METHODS, TOOLS, TECHNOLOGIES, AND TECHNIQUES

4.1 Synthetic Data Generation and Feature Engineering

To generate training data that closely mirrors real-world medical datasets, we developed a sophisticated synthetic data generation pipeline. Recognizing that actual patient data exhibits significant variability and includes extreme outliers, we employed advanced statistical distributions rather than simple normal distributions. For continuous features such as glucose level, blood pressure, and BMI, we implemented composite distribution approaches. Glucose levels were modeled using a combination of normal distribution (mean=100, SD=30) and exponential distribution ($\lambda=50$) to simulate both typical patients and diabetic extremes, with values clipped between 60-450 mg/dL. Blood pressure incorporated a gamma-distributed stress factor ($k=2$, $\theta=10$) added to a base normal distribution (mean=120, SD=15), clipped between 90-240 mmHg to represent hypertension crises. BMI employed a beta distribution ($\alpha=2$, $\beta=5$) scaled to the range 15-60, producing the right-skewed distribution characteristic of real population data, where most individuals fall within normal ranges but a tail represents severely obese patients. Age range was expanded from 15-90 to 15-95 years to capture elderly high-risk populations. This approach ensures our models are trained on data that

authentically represents both typical cases and critical outliers they will encounter in clinical deployment.

4.2 Score Calculation and Normalization Strategy

Our initial approach employed cumulative distribution function (CDF) normalization to map raw disease scores to percentile values. However, we identified a fundamental limitation: CDF assumes normally distributed input data, which proved problematic given the skewed nature of our synthesized medical features. This normalization method compressed extreme risk cases into the 70-80 percentile range rather than properly reflecting their critical severity. Consequently, we transitioned to min-max scaling, which preserves the full range of risk scores without distributional assumptions. Under this approach, a patient with extremely elevated risk factors now receives scores approaching 100, while low-risk patients receive scores near 0, maintaining the true severity gradient essential for clinical decision-making. The formula applied is: $\text{normalized_score} = (\text{raw_score} - \text{min_score}) / (\text{max_score} - \text{min_score}) \times 100$, ensuring scores span the complete 0-100 range and accurately reflect relative risk levels.

4.3 Hierarchical Two-Layer Prediction Architecture

We implemented a hierarchical two-layer machine learning architecture designed to balance computational efficiency with prediction accuracy. The first layer consists of a lightweight screening model (`model_important`) trained on 12 key features common across all three diseases, providing rapid initial risk assessment for stroke, diabetes, and hypertension simultaneously. This model processes all incoming patients and generates preliminary percentile scores. Patients whose scores exceed dynamically calculated disease-specific thresholds are then forwarded to the second layer: specialized deep-analysis models (`model_stroke`, `model_diabetes`, `model_hypertension`) trained on comprehensive disease-specific feature sets. For example, the stroke-specific model utilizes all stroke-related features including `paralyzed_area`, `speech_difficulty`, and `balance_issues`—features not present in the initial screening. This architecture ensures that the majority of low-risk patients receive fast, efficient screening, while high-risk cases benefit from detailed specialist evaluation without overwhelming the system with unnecessary deep computations for every patient.

4.4 Dynamic Threshold Calibration Methodology

Rather than arbitrarily selecting threshold values for routing patients to specialized models, we developed a data-driven threshold calibration method. Using a reference model trained on all 27 features (`model_all`) as ground truth, we systematically evaluate the screening model's performance across the full percentile range (0-100) in 1-percentile increments. For each threshold candidate, we calculate the recall (percentage of truly high-risk patients correctly flagged) and the resulting patient load (percentage of total patients sent to specialists). The optimal threshold is determined by identifying the point where recall first exceeds 95%, ensuring we capture the vast majority of at-risk individuals while minimizing unnecessary specialist referrals. This approach resulted in empirically determined thresholds: stroke $\approx$ 40-45%, diabetes $\approx$ 35-40%, and hypertension $\approx$ 45-50%. The calibration process is visualized through threshold analysis plots showing the trade-off between patient safety (high recall) and system efficiency (manageable load). (see Appendix I)

4.5 XGBoost Model Implementation and Training

All models in our pipeline employ XGBoost Regressor (`XGBRegressor`) configured for continuous percentile prediction rather than binary classification. This choice was motivated by the need for nuanced risk stratification — we aim to distinguish between a patient at 45% risk versus 75% risk, not merely "high risk" or "low risk." The models were trained on synthetically generated datasets: `model_all` uses `csv_all_v2` (all 27 features), `model_important` uses `csv_important_v2` (12 core features), and disease-specific models use `csv_stroke_v2`, `csv_diabetes_v2`, and `csv_hypertension_v2` respectively. Each dataset

contains synthetic patient records with disease likelihood percentiles as regression targets. Feature scaling was performed separately for each model using StandardScaler to ensure numeric features (glucose_lvl, blood_p) are normalized, while binary categorical features remain unchanged. The complete prediction pipeline integrates these components: patient data → scaling → screening model → threshold comparison → conditional routing to specialist models → final refined percentiles. The technical implementation across all models was done in Python 3.10 using Pandas and NumPy for data processing, Scikit-learn for metrics, calibration, and preprocessing, XGBoost for building the main models, Matplotlib and Seaborn for visualizations, and Joblib for saving model artifacts and ensuring reproducibility.

4.6 Diabetes Risk Prediction Model

To develop our diabetes risk prediction model, we used a standard structured pipeline consisting of data preprocessing, feature engineering, model building, probability calibration, quality assessment, and interpretability analysis. We used the open-source BRFSS 2015 (Behavioral Risk Factor Surveillance System) dataset from Kaggle (Soriano, n.d.) which is one of the world's largest medical survey databases, including more than 250,000 individual questionnaires with a wide range of demographic, behavioral, social, and easily measurable medical indicators at home. In the first step, we carefully checked all features for missing values, incorrect values, and coding inconsistencies. Depending on the type of variable, missing values were either deleted or corrected by imputation. Most BRFSS features are presented in text or categorical form, so we converted them to numerical format using one-hot encoding and checked that all the necessary parameters were present in our processed dataset. To handle the pronounced class imbalance in the dataset, we used the `scale_pos_weight` parameter of the XGBoost algorithm, which increased the weight of positive examples, thereby allowing the model to more effectively recognize rare cases of diabetes. In addition, irrelevant features were removed from the dataset, and numerical variables were standardized to increase the stability of the training process. We created a binary version of the target variable, `Diabetes_binary`, where all cases with 1 and 2 are combined into the "diabetes" group. Appendix B (see Appendix B). During the model development process, we selected hyperparameters such as learning rate, tree depth, number of estimators, and stratified cross-validation to ensure the reliability of estimates and avoid overfitting. Initial results showed a discrepancy between the predicted probabilities and the actual frequencies of events, which was expected with a highly imbalanced dataset. To address this issue, we applied `CalibratedClassifierCV`, and the quality of the calibration was evaluated using calibration curves, reliability charts, and Brier Score. After calibration, probability thresholds were developed to stratify users into three risk groups: low, medium, and high. The thresholds were calculated based on the analysis of Precision–Recall curves, F1 metric optimization, and clinical requirements that prioritize reducing the probability of missing potentially ill patients. To evaluate the quality of the model, our team used a wide range of metrics relevant to class imbalance tasks, with key focus on ROC-AUC, PR-AUC, and Brier Score, as well as precision, recall, and F1-score indicators for each class. We also created visualizations, including ROC curves, Precision–Recall diagrams, and calibration graphs, to provide a deeper understanding of the model's behavior.

4.7 Hypertension Risk Prediction Model

We selected XGBoost for this model due to its strong performance on non-linear lifestyle factor relationships. Before training, we spent a good amount of time on the preprocessing dataset to make sure the model remained clinically meaningful. For example, we removed features like "BP_History" and "Medication" because they basically gave away the answer and wouldn't be available in a real pre-diagnosis setting. Also we have simplified some inputs so they would be easier for users to provide. One example is daily salt intake: instead of keeping it as a raw number, we've converted it into Low, Moderate, or High ranges using percentiles that matched typical consumption patterns. For the actual ML preparation, we used a mix of encoding and scaling. Ordinal features, like exercise level, were mapped to

numbers so the model would understand their order. For numerical columns, StandardScaler was applied, since features like Age and Stress Score had very different scales and would otherwise affect the model unevenly. After preparing everything, we trained an XGBoost classifier. Even though the dataset was mostly balanced, we calculated the `scale_pos_weight` parameter just to correct any slight imbalance and avoid the model leaning towards the majority class. Then we tuned the model using GridSearchCV with 5-fold cross-validation, which was mainly focused on parameters like tree depth, learning rate, subsampling ratios, and optimized the F1-Score so the model wouldn't favor either Precision or Recall too much. Finally, instead of using a simple positive/negative output, our team has built a three-level risk system. By looking at the Precision-Recall Curve, we picked two probability cutoffs: one where we could be about 90% sure a person is healthy, and another where we had similarly high confidence in a positive case. Predictions that fell in between those two thresholds were labeled medium risk.

4.8 Stroke Risk Prediction Model

We developed a complete machine-learning pipeline to predict stroke risk using a publicly available dataset from Kaggle. During our exploratory data analysis, we identified missing values, outliers, and severe class imbalance as key quality issues. After eliminating the duplicated rows to not harm our models prediction power, we imputed numerical gaps using the mode/median, and we transformed categorical variables via one-hot encoding. We also added SMOTE to address class imbalance. The next step was converting all the features into numerical features and for this we applied StandardScaler because of its outlier scaling power. Then our team partitioned the dataset into training, validation, and test sets, which allowed us to use stratified sampling to preserve class proportions. We used XGBoost with the `scale_pos_weight` parameter to further control class imbalance. To improve the reliability of probability estimates, we applied the same type of calibration. Further ROC and Precision–Recall analyses were used to determine optimal decision thresholds, enabling us to construct a three-level risk system (green, yellow, red) based on calibrated probabilities. We subsequently implemented a threshold-based scoring mechanism, and we conducted feature-importance analyses to identify the most influential predictors in the model.

4.9 Sample Size Estimation

In this study, we use a structured online questionnaire to collect demographic and contextual indicators of patients in order to develop machine learning models for predicting the risk of chronic diseases such as hypertension, diabetes, and stroke. As a survey tool, we used the Google Form, which is structured in such a way as to obtain binary and categorical results related to the prevalence of diseases and risky behaviors.

The sample size indicates the number of people who need to be interviewed so that the results are representative of the entire target patient population and with the required degree of accuracy. For finding a sample in medical research while estimating the proportion of the population in survey-based studies, if the exact number of patients with certain diseases is not known, we will use the classical approach by using Cochran's formula.

$$n = \frac{Z^2 * p(1 - p)}{d^2}$$

Where:

n = required sample size

Z = Z-score corresponding to the selected confidence level

p = expected population proportion

d = margin of error

Also before using this formula several key parameters must be considered:

- Confidence level represents the degree of certainty that the true population parameter lies within the calculated confidence interval. From **S. Charan and T. Biswas'** publication we pointed that in biomedical research, a 95% confidence level is most commonly adopted.
- Margin of error defines the maximum acceptable deviation between the estimated sample proportion and the true population proportion. As **Elmas Pinar Kahraman Kılbaş and Fatma Cevahir** said in their work that 5% margin of error is acceptable in social research.
- Expected proportion defines the expected distribution of a characteristic of interest (presence of a disease in our case).

Usually, in order to obtain statistically significant results for each disease, the sample is calculated for each disease separately. As part of determining the sample size, we can use current epidemiologic data on the prevalence of the studied diseases in Turkey for the last 10 years. According to current national studies, the prevalence of arterial hypertension in the adult population is approximately 30-33%, diabetes is 16-17%, and stroke is approximately 1-1.3% of the adult population. By knowing all of this parameters we can calculate sample sizes:

•If $p = 0.5$:

$$n = \frac{1.96^2 * 0.5(1-0.5)}{0.05^2} = 384.16 \approx 385$$

So, we will need to interview about 385 patients for each of the three diseases. That is, about 1,155 questionnaires in total.

•For diversity p :

Diabetes ($p = 0.11$)

$$n = \frac{1.96^2 * 0.11(1-0.11)}{0.05^2} = 150.72 \approx 151$$

Hypertension ($p = 0.3$)

$$n = \frac{1.96^2 * 0.3(1-0.3)}{0.05^2} = 322.69 \approx 323$$

Stroke ($p = 0.01$)

$$n = \frac{1.96^2 * 0.01(1-0.01)}{0.05^2} = 15.21 \approx 16$$

For stroke, the number turned out to be small (16), because the formula evaluates the accuracy of the prevalence estimate, rather than sufficiency for ML. But to build a model, you need to get at least 200 stroke patients with $p = 1\%$: $200 / 0.01 = 20,000$ people. That's why rare diseases require a huge sample. It is also worth considering the fact that in practice not all patients complete the questionnaire, and some data may be lost. In clinical trials, it is strongly recommended to add a buffer of 10-20% to the initially calculated number. Then sample sizes will be:

- $p = 0.5$: $385 + 10\% = 424$ for each of diseases
- Diabetes ($p = 0.11$): $151 + 10\% = 166$
- Hypertension ($p = 0.3$): $323 + 10\% = 356$
- Stroke ($p = 0.01$): $16 + 10\% = 18$

To ensure that the study sample was representative of the target population and to achieve the specified precision, sample sizes were also calculated using the G*Power software, as this program is most commonly used in modern medical research. We conducted a prior power analysis using a family of exact tests, with the initial parameters set at 95% statistical power ($1-\beta = 0.95$) and a 5% alpha error probability ($\alpha = 0.05$) (**see Appendix L**). By entering the expected proportions of the national population for each specific disease along with the corresponding effect sizes, the initial required sample sizes were calculated: 1,143 for

hypertension (fixed proportion $p = 0.30$, effect size $g = 0.05$), 601 for diabetes ($p = 0.11$, $g = 0.05$), and 1,857 for stroke ($p = 0.01$, $g = 0.01$). Given that in practice not all patients complete the questionnaires and some data may be lost, a 10% margin of error was added to these initial calculations to avoid statistical underestimation. Ultimately, this robust methodological approach allowed us to determine the final required sample sizes: 1,258 for hypertension, 662 for diabetes, and 2,043 for stroke. (see **Appendix L**).

4.10 First part of algorithmic optimization of clinical routing pathways

Our routing system is implemented in Python using the NumPy and Pandas libraries. NumPy processes numerical arrays and enables fast vector computations, while Pandas is used for sorting and filtering the resulting data on hospital distribution.

We treat patients and hospitals as object classes containing x and y geographic coordinates. The initial distance between a patient and each hospital is calculated using the fast Euclidean norm formula: $\text{dist_km} = \sqrt{\text{locationpatient} - \text{locationhospital}} \times 111$. We multiply by 111 specifically because this is the method used to quickly convert map coordinates from degrees to kilometers, providing an approximation that is both computationally fast and highly accurate for modeling urban areas. To ensure that distance does not mathematically overshadow other critical factors, such as hospital capacity, the adjusted distance is normalized. Our algorithm also divides the adjusted distance by $\text{MAX_DIST_ANKARA} = 70$ km, which corresponds to the approximate diameter of the city, to scale it to a range from 0 to 1. The final influence of distance on the algorithm's route decision is determined by a dynamic weight variable that changes depending on the urgency of the patient's condition:

- **Red (Critical/Stroke):** Distance is assigned a 95% weight ($w = 0.95$), making physical proximity the overwhelming priority to satisfy the "Golden Hour" principle.
- **Yellow (Urgent but stable):** Distance is assigned a 60% weight ($w = 0.60$), mathematically balancing the travel distance with the expected hospital waiting time.
- **Green (Low risk):** Distance is given a minimal 25% weight ($w = 0.25$), allowing the algorithm to prioritize sending the patient to a hospital with a lighter workload rather than the closest one.

4.11 Ethics Approval Process and Survey Design Methodology

A key limitation of our current approach is that the three disease prediction models were trained on separate, independent datasets with no patient overlap, meaning the system cannot yet assess the same individual for all three diseases using real-world data. To address this, we designed a unified online questionnaire that consolidates all relevant variables from the diabetes, hypertension, and stroke datasets into a single instrument, enabling cumulative data collection and real-world validation of our predictions. Since the questionnaire collects personal health information, we applied to the Ankara Yıldırım Beyazıt University Health Sciences Ethics Committee (Sağlık Bilimleri Etik Kurulu) for approval before beginning data collection. The ethics application included informed consent procedures, participant anonymization through randomly assigned IDs, voluntary participation conditions, and a five-year encrypted data retention policy. The survey was implemented using Google Forms and comprises 54 questions organized into seven sections covering all feature variables required by our three prediction models.

5. DOCUMENTATION OF ALL TASKS ACCOMPLISHED

5.1 Diabetes Risk Prediction Model

We conducted a literature review identifying the gap in non-laboratory diabetes prediction models, which justified our selection of the BRFSS 2015 dataset containing over 250,000 observations with demographic, behavioral, and self-reported health indicators (see **Appendix A.1**). After preprocessing — removing irrelevant features, handling missing values, converting categorical variables to numerical format, and creating a binary target variable — we performed exploratory data analysis that confirmed class imbalance and

identified key risk factor relationships (**see Appendix B**). We trained an XGBoost model with `scale_pos_weight` to handle class imbalance, optimized hyperparameters through 5-fold cross-validation, and saved the final model as an artifact (**see Appendix C**). Probability calibration was performed using `CalibratedClassifierCV`, and optimal risk stratification thresholds were determined from the Precision–Recall curve (**see Appendix D, J.1**). Final evaluation metrics including ROC-AUC, PR-AUC, F1-score, and confusion matrices are presented in (**Appendix E, F**).

5.2 Hypertension Risk Prediction Model

After reviewing multiple candidate datasets, we selected the hypertension-risk-prediction-dataset (Miadul) as it best aligns with our objective of using only self-identifiable contextual data (**see Appendix A.2**). We removed leakage-prone features (BP_History, Medication), standardized categorical fields, and transformed continuous salt intake into percentile-based categories. EDA confirmed a fairly balanced dataset (52% positive) with low multicollinearity (**see Appendix G**). We trained an XGBoost classifier with `scale_pos_weight` = 0.92, tuned via `GridSearchCV` with 5-fold cross-validation, achieving 78% accuracy and 76% recall for the positive class. A three-tier risk system was built using Precision–Recall curve analysis to determine probability cutoffs (**see Appendix H, J.2**). The final model was saved in `pkl` format for integration into the two-level pipeline.

5.3 Stroke Risk Prediction Model

We selected the Stroke Prediction Dataset from Kaggle (**see Appendix A.3**), which exhibited significant class imbalance (positive class < 5%) and missing BMI values. After preprocessing — median imputation by age group, one-hot encoding of categorical features, and stratified train-test splitting — we performed EDA confirming the strong influence of age, hypertension, heart disease, and glucose levels (**see Appendix I**). SMOTE was applied to the training set, and an XGBoost classifier was trained with tuned hyperparameters. Probability calibration and Precision–Recall analysis produced a three-level risk stratification system (**see Appendix J.3**). Despite overall accuracy exceeding 90%, recall and precision for the positive class remain low, and further improvement is planned (**see Appendix K**).

5.4 Advanced Synthetic Data Generation System

We successfully developed and implemented a comprehensive synthetic data generation pipeline (`create_csv_files.ipynb`) that produces realistic training datasets. The pipeline transforms medical knowledge encoded in feature tables into 10 distinct CSV files containing synthetic patient records with disease percentile labels. (**see Appendix M**) We conducted two iterations: version 1 utilized basic normal distributions as a proof-of-concept, while version 2 incorporated sophisticated statistical approaches (exponential, gamma, and beta distributions) to authentically replicate real-world medical data characteristics including extreme outliers. This advancement ensures our machine learning models encounter diverse patient profiles during training, including critical edge cases they will face in clinical deployment.

5.5 Implementation of Hierarchical Prediction Models

We designed and trained a two-layer machine learning architecture consisting of six XGBoost regression models: one reference model (`model_all`) using all features for validation purposes, one rapid screening model (`model_important`) for initial assessment, and three specialized models (`model_stroke`, `model_diabetes`, `model_hypertension`) for detailed risk analysis. Each model was trained on appropriately structured synthetic data and validated to ensure reliable percentile predictions. The hierarchical design optimizes both computational efficiency and clinical accuracy by reserving deep analysis for patients flagged by the screening layer.

5.6 Development of Dynamic Threshold Calibration

We created an automated threshold calibration system that determines optimal routing criteria between screening and specialist models. By systematically analyzing model performance across all percentile values, we established disease-specific thresholds that achieve $\geq 95\%$ recall while maintaining manageable referral loads. This data-driven approach eliminates arbitrary threshold selection and ensures the system dynamically adapts based on actual model behavior and clinical priorities. (see **Appendix N**)

5.7 Sample Size Estimation

As part of the project, our team developed a comprehensive data collection methodology, designing a structured online questionnaire in Google Forms that effectively gathers demographic data and specific risk markers needed to train machine learning models aimed at predicting hypertension, diabetes, and stroke. When determining the statistical parameters ensuring the representativeness of the sample, we found national scientific articles in which experts established a 95% confidence level and a 5% margin of error; while, by departing from standard conservative assumptions, they successfully integrated current epidemiological data on Turkey into the calculations, according to which the prevalence of hypertension is 30–33%, diabetes 11–12%, and stroke 1–1.3%. The process of calculating the sample size involved both the classical application of Cochran's formula and an in-depth power analysis using G*Power software (see **Appendix L**), which results we decided to use, where, relying on logistic regression and tests of proportion differences, the specialists determined baseline quantitative benchmarks for each condition. In the final stage, taking into account the inevitable risk of incomplete questionnaires and seeking to minimize the impact of human error, the team applied a 10% adjustment for potential data loss, which allowed them to finalize the sample sizes: 1,258 respondents for the analysis of hypertension, 662 for diabetes, and 2,043 for the study of stroke.

5.8 First part of algorithmic optimization of clinical routing pathways

According to the project's task completion documentation, the main results related to patient routing based on distance include several key steps. First, we identified the factors that influence patient distribution, and one of the primary parameters was the distance between the patient and the hospital. Next, we normalized and weighted this distance along with hospital capacity levels so that the final decision would account not only for proximity but also for the actual capabilities of medical facilities, with the weights dynamically adjusting based on the patient's risk level. We also implemented a backup route system: if suitable specialized hospitals are unavailable, the system automatically selects the nearest facility by distance as an emergency option. Finally, we added a reporting feature that records all calculated distances and final hospital ratings. During the development process, we reviewed existing research and found that traditional approaches rely primarily on distance alone and ignore factors such as load and external conditions, whereas modern methods employ more comprehensive models that incorporate multiple criteria. Based on this, we expanded our approach: now the system doesn't just calculate distance, but can adjust it based on weather conditions and traffic, normalizes values within constraints, and then makes a decision based on medical prioritization or, in emergency cases, simply selects the nearest hospital as the most reliable option.

5.9 Ethics Committee Approval and Survey Design

We prepared and submitted an ethics application to the Ankara Yıldırım Beyazıt University Health Sciences Ethics Committee on 9 January 2026 under the research title "Yapay Zeka Temelli Sağlık Riski Değerlendirmesi ve Klinik Kaynak Yönetimi" (Research Code: 26). The submission included documentation of study objectives, informed consent procedures, anonymization measures, and data retention policies. The application was approved on 22 January 2026 (Decision No: 01/26) as ethically appropriate (see **Appendix O**). Following approval, we designed a structured 54-question Google Forms questionnaire beginning with

an Informed Voluntary Consent Form and organized into seven thematic sections: demographic and socio-economic variables, physical measurements, lifestyle and habits, medical history and diagnoses, family history, general health and psychological status, and symptoms and signs. All questions collect binary and categorical responses aligned with the non-laboratory indicators used in our machine learning models. The questionnaire is currently being distributed to collect real-world validation data (**see Appendix P**).

6. INITIAL FINDINGS, PROTOTYPES, OR OUTPUTS OBTAINED THUS FAR

So far, we have built and tested early prototypes for three chronic diseases. The system operates through a two-stage ML pipeline: a multi-class XGBoost model trained on 30,000 synthetic records identifies the patient's dominant pathology, followed by disease-specific models that calculate personalized risk scores. The diabetes model uses XGBoost with CalibratedClassifierCV for 0–100% risk scoring, the stroke model uses an ANN optimized for high recall, and the hypertension model uses Logistic Regression with calibrated probability thresholds. Results are communicated through a three-tier system: Green (no significant risk), Yellow (behavioral risk patterns), and Red (critical probability requiring intervention). Ethical standards have been confirmed by the Ethics Committee's approval.

We improved the ML pipeline by implementing more realistic synthetic data distributions with outliers, transitioning from CDF to Min-Max normalization, adopting a two-tiered model architecture, and adding dynamic threshold calibration. Validation was conducted using three test categories: edge cases (unambiguous single-disease dominance), close cases (overlapping symptoms), and adversarial cases (deliberately misleading labels). The models demonstrated robust performance across all categories, consistently following actual data patterns rather than erroneous labels.

To facilitate real-world validation, a structured online questionnaire was developed with Ethics Committee approval, and sample sizes were statistically determined using G*Power software based on national prevalence rates in Turkey, with a 10% buffer for data loss (**see Appendix L**). The practical significance of the work lies in increasing patient survival through preventive monitoring and optimizing emergency medical services.

In parallel with the model development work, we completed the ethical approval process required for collecting real-world patient data. The ethics application was approved on 22 January 2026 by the AYBÜ Health Sciences Ethics Committee (Research Code: 26, Decision No: 01/26) (**see Appendix O**). Following approval, we designed a unified 54-question Google Forms questionnaire that consolidates variables from all three disease datasets into a single survey instrument, currently being distributed for real-world data collection (**see Appendix P**).

We also developed a comprehensive system flowchart illustrating the end-to-end pipeline, depicting two parallel paths — Path A (Personal) for user input through L1 screening and L2 disease-specific risk scoring, and Path B (Sample Cases) for pre-filled demonstration data — both converging at the hospital routing stage where a ranking algorithm based on distance, capacity, and risk match generates hospital recommendations (**see Appendix Q**).

7. SCHEDULE REFLECTING COMPLETED AND REMAINING TASKS

Phase	Activities	Duration	Deliverables	Execution Status
Phase 1: Preparation & Data Collection	- Conduct in-depth literature review - Identify, collect, and consolidate open-access datasets (UCI, Kaggle, WHO) for diabetes, hypertension, and stroke. - Perform data preprocessing (cleaning, normalization) and Exploratory Data Analysis (EDA).	17.10.2025 - 27.11.2025	- Curated and preprocessed datasets. - An EDA and feature importance report.	Completed
Phase 2: Machine Learning Model Development	- Develop the two-stage ML model: Stage 1: Train models to predict the individual risk score/severity for each disease. Evaluate and compare model performance (Accuracy, Precision, Recall, F1-score, ROC-AUC) to select the best models. Stage 2: Train classifier to determine the most likely disease	28.11.2025 - 22.01.2026	- Trained and validated ML models for disease classification and risk scoring. - A model evaluation report justifying the chosen algorithms.	Completed
Phase 3: Optimization Model Development	- Design and develop the optimization algorithm for hospital resource allocation . - Create a simulation environment with virtual hospitals and varying capacities . - Integrate the validated ML models and the optimization algorithm	23.01.2026 - 12.03.2026	- A functional optimization algorithm. - A prototype of the integrated web application. - Initial simulation results for the optimization algorithm.	In progress
Phase 4: Web Application Development	-Development of the user-friendly web application .	13.03.2026 - 23.04.2026	Completion of a functional web prototype that allows the user to	Not processed

Phase	Activities	Duration	Deliverables	Execution Status
and Module Integration	-Integration of ML models and the optimization module into a web-based decision support system and design of the user interface.		enter data and obtain the risk score and simulation result.	
Phase 5: Validation, Refinement, & Final Reporting	- Conduct final system-wide evaluation and simulation experiments to compare with conventional methods . - Finalize and deploy the web application	13.03.2026 - 23.04.2026	- Final research paper and presentation. - The completed web-based decision support system.	Not processed

8. OBSTACLES ENCOUNTERED AND HOW THEY WERE OR WILL BE ADDRESSED

As we know and have seen in earlier scientific works, the development of machine learning systems for medicine is inevitably accompanied by a whole range of difficulties related to both the quality of data and the specifics of its processing, interpretation, and implementation in practical use. However, we would like to note that their timely identification and resolution became an important part of the research methodology, since the quality of the final model directly depends on the correctness of the work at all stages from data cleaning to the formation of risk score prediction pipeline. The following subsection describes in detail the key problems that arose during the work on each of the models.

8.1 Diabetes Risk Prediction Model

One of the most significant problems was the strong class imbalance in the BRFSS dataset, where the number of diabetes cases was significantly lower than the number of negative examples, which in turn led to low completeness and accuracy values for the minority class when building base models. To solve this problem, we used the `scale_pos_weight` parameter in the XGBoost algorithm, which compensated for the unevenness of the classes. We calculated `scale_pos_weight` using the formula: number of negative examples / number of positive examples. This solution increased the contribution of rare positive observations to the loss function, thereby improving the model's ability to correctly identify cases of diabetes. The main quality indicators used were PR-AUC and recall metrics, which reflect the model's ability to detect rare positive cases. The application of this mechanism significantly improved the Recall and PR-AUC metrics compared to unadjusted models. In addition, we used probability calibration and classification threshold optimization, which improved the detection of high-risk users. Unfortunately, the second obstacle was that the BRFSS dataset contains a large number of categorical and ordinal variables, as well as non-standard formats and gaps, which complicated the preprocessing stages and forced us to ensure strict data consistency. To address this issue, we used data preprocessing methods that ensure stable conversion of features into numerical form. This data processing approach was used both during training and prediction to eliminate discrepancies between stages. The third problem was the low calibration of probabilities in the early stages of development. Although the basic models could provide acceptable classification accuracy, the probability distribution did not correspond to the actual level of risk, which significantly limited the ability to divide users into low, medium, and high risk categories. To address this issue, we applied the

CalibratedClassifierCV isotonic method, which allowed us to obtain more reliable probability estimates for nonlinear relationships between features, and the improvements were confirmed by analyzing calibration curves and a sufficiently high Brier Score metric, which shows how accurately the probabilities were found. Next obstacle concerned computational costs, since training XGBoost on a large number of features and data volumes required increased time and resources. However, to optimize the process, training was initially conducted on small subsets, which allowed us to quickly select suitable hyperparameters, and full experiments were launched only after preliminary analysis. The last difficulty may be related to the integration of the model into a two-level forecasting system, where it will be necessary to ensure consistency between the training stage, input data format, and final predicting function. However, the model and related settings were stored in a structured package, and we hope that this will make it easy to use them in the future when creating a user interface.

8.2 Hypertension Risk Prediction Model

During the development of the Hypertension Risk Prediction model, we ran into several data and modeling obstacles that we had to fix to keep the system clinically meaningful. One of the first problems came up during the exploratory analysis: data leakage. A couple of features, especially *BP_History* and *Medication*, were basically giving away the answer because they reflect the outcomes of hypertension rather than the factors that lead to it. If we left them in, the model would look extremely accurate on paper but completely useless for screening people who haven't been diagnosed yet. To prevent this, we added a strict leakage-removal step in the preprocessing data, making sure the model only learned from lifestyle and behavioral features that could actually be measured before diagnosis. We also faced a practical issue with some continuous variables—daily salt intake was the biggest one. It's not realistic to expect normal users to know exactly how many grams of salt they consume on average everyday. To make the model usable outside of a controlled research environment, we transformed this variable using percentile-based bins. By taking the 33rd and 67th percentiles from the training set, we created three categories (Low, Moderate, High) that still preserved meaningful statistical differences while allowing users to give simple, categorical answers. On the modeling side, the dataset size (around 2,000 entries) showcased a big overfitting risk, especially with a model like XGBoost. Using the default settings would most likely cause the model to memorize noise. To avoid that, we ran GridSearchCV with 5-fold cross-validation and intentionally restricted the model's flexibility—setting the max tree depth to 3 and lowering the learning rate to 0.05. This kept the model from fixating on small quirks in the data and encouraged it to learn broader patterns. We also had to think about the medical side of the problem. Standard metrics like accuracy treat False Negatives and False Positives the same, which doesn't make sense in a health screening context, because missing an at-risk patient is far worse than flagging a healthy person for follow-up. To handle this, we adjusted the `scale_pos_weight` parameter so the model would pay more attention to the positive (at-risk) class. Then we went beyond the default 0.5 probability threshold and looked at the Precision–Recall Curve to find a safer cutoff, that increases Recall and makes sure that medium-risk patients are flagged earlier and in this way we reduce the chance of overlooking someone who might actually be high-risk.

8.3 Stroke Risk Prediction Model

The most critical challenge was the extreme class imbalance in the stroke dataset. This caused our initial models to strongly favor the majority class, resulting in near-zero recall (sensitivity) for actual stroke cases. We fixed this by using class-weighted training and stratified splitting when preparing the data. While these steps helped, future work may still benefit from advanced techniques like SMOTE or hybrid oversampling to improve sensitivity further. Also we found that running probability calibration improved the reliability of our risk scores but unexpectedly suppressed the prediction of positive cases. To solve this, we customized the probability thresholds based on the Precision-Recall curve, moving away from the standard 0.5 cutoff to optimize for safety to save high recall. We also ensured data

fidelity by retaining categorical variables with large "Unknown" categories (like smoking status) and applying one-hot encoding, rather than dropping the data and distorting the dataset's overall patterns. Although certain feature distributions, such as age and glucose, were highly skewed, the chosen XGBoost algorithm successfully managed these distributions without needing complex transformations. Currently, despite the fact that the model's accuracy is very high, over 90%, we decided to continue working on this model and search for new algorithms for classifying patients, since the recall and precision for the patient class are very low, which is clinically incorrect. Our next steps will focus on implementing advanced methods for eliminating imbalance and conducting external validation to ensure effective model generalization.

8.4 Ethics Approval and Survey Data Collection

One of the major challenges we encountered was the requirement to obtain Ethics Committee approval before distributing our survey, as the questionnaire collects personal health information from participants. The application process required preparing extensive documentation including informed consent forms, anonymization procedures, data retention policies, and detailed descriptions of the research methodology, all of which had to comply with the committee's strict regulations. This process introduced a delay in our data collection timeline, as the application was submitted on 9 January 2026 and the committee review was completed on 22 January 2026. Another challenge was designing a single unified questionnaire that consolidates the feature variables from three separate disease datasets into one coherent survey instrument. Each disease model was trained on a different dataset with different variable structures, so aligning the questions to capture all necessary inputs while keeping the survey accessible and understandable for non-medical respondents required careful balancing. We addressed this by organizing the 54 questions into seven thematic sections that follow a logical flow from general demographics to disease-specific symptoms, ensuring completeness without overwhelming the participant. Additionally, the required sample sizes calculated through our power analysis, particularly for stroke (2,043 respondents due to its low prevalence of approximately 1%), present an ongoing challenge in terms of reaching a sufficient number of respondents to validate our models with statistical significance.

9. OUTLINE OF PENDING TASKS AND MILESTONES FOR THE REMAINDER OF THE PROJECT

- Milestone 1: Compilation of three separate models, each designed for a different disease. These models will be tested and refined as needed to achieve reliable performance and can be improved by using other ML, NN models or AI algorithms.
- Milestone 2: Development of the second-level machine learning model, whose purpose is to identify the most likely disease. The disease predicted by this model will then be evaluated using the corresponding disease-specific ML model to ensure optimal accuracy.
- Milestone 3: Creation of a simulation environment featuring virtual hospitals with varying capacities. This environment will be used to simulate potential patient allocation scenarios.
- Milestone 4: Development of a user friendly web application. The ML models and the optimization algorithm will be integrated into this platform to deliver a complete and practical end-to-end solution.
- Milestone 5: Conduct final system-wide evaluation and simulation experiments to measure model accuracy, distribution balance, and decision support effectiveness across various patient scenarios, followed by the preparation of the final research paper and project presentation.

APPENDICES

Appendix A --- Chosen datasets for each disease's model

```

General Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 253680 entries, 0 to 253679
Data columns (total 22 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Diabetes_012          253680 non-null float64
1   HighBP                253680 non-null float64
2   HighChol              253680 non-null float64
3   CholCheck             253680 non-null float64
4   BMI                   253680 non-null float64
5   Smoker                253680 non-null float64
6   Stroke                253680 non-null float64
7   HeartDiseaseorAttack  253680 non-null float64
8   PhysActivity          253680 non-null float64
9   Fruits                253680 non-null float64
10  Veggies               253680 non-null float64
11  HvyAlcoholConsump     253680 non-null float64
12  AnyHealthcare         253680 non-null float64
13  NoDocbcCost           253680 non-null float64
14  GenHlth              253680 non-null float64
15  MentHlth              253680 non-null float64
16  PhysHlth              253680 non-null float64
17  DiffWalk              253680 non-null float64
18  Sex                   253680 non-null float64
19  Age                   253680 non-null float64
20  Education              253680 non-null float64
21  Income                253680 non-null float64
dtypes: float64(22)
memory usage: 42.6 MB

```

Figure 1 Chosen diabetes BRFSS 2015 dataset

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1985 entries, 0 to 1984
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   1985 non-null   int64
1   Stress_Score          1985 non-null   int64
2   BP_History            1985 non-null   int64
3   Sleep_Duration        1985 non-null   float64
4   BMI                   1985 non-null   float64
5   Family_History         1985 non-null   int64
6   Exercise_Level        1985 non-null   int64
7   Smoking_Status        1985 non-null   int64
8   Has_Hypertension      1985 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 139.7 KB

```

Figure 2 Chosen hypertension-risk-prediction-dataset (Miadul) hypertension dataset

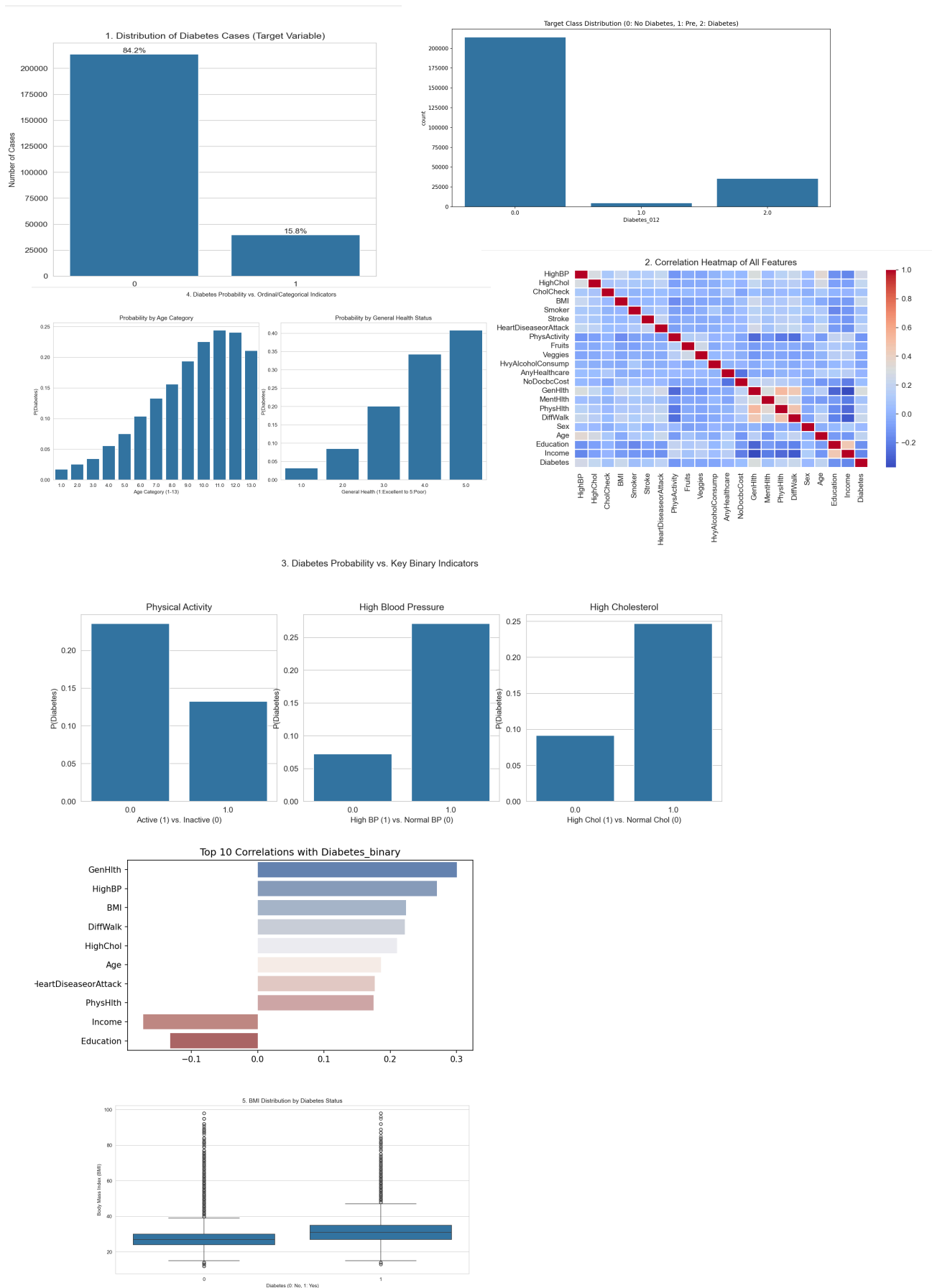
```

General Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                    5110 non-null   int64
1   gender                5110 non-null   object
2   age                   5110 non-null   float64
3   hypertension          5110 non-null   int64
4   heart_disease         5110 non-null   int64
5   ever_married          5110 non-null   object
6   work_type             5110 non-null   object
7   Residence_type        5110 non-null   object
8   avg_glucose_level     5110 non-null   float64
9   bmi                   4909 non-null   float64
10  smoking_status        5110 non-null   object
11  stroke                5110 non-null   int64
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB

```

Figure 3 Chosen stroke “healthcare-dataset-stroke-data.csv” dataset

Appendix B --- EDA for diabetes BRFSS 2015 dataset



Appendix C --- Structure of the trained XGBoost and model's saved artifacts

```

sick = (y_tr == 0).sum()
healthy = (y_tr == 1).sum()
# balance the classes
scale_pos_weight = max(1.0, sick / max(1, healthy))

xgb = XGBClassifier(
    n_estimators=800,
    learning_rate=0.05,
    max_depth=5,
    subsample=0.9,
    colsample_bytree=0.9,
    min_child_weight=5,
    reg_lambda=1.0,
    random_state=42, n_jobs=-1,
    objective="binary:logistic",
    eval_metric="auc", tree_method="hist",
    scale_pos_weight=scale_pos_weight)

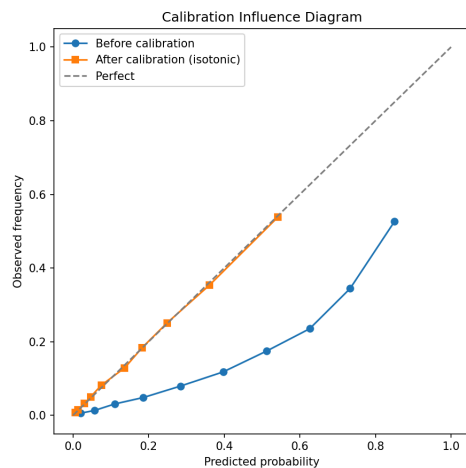
```

```

artifacts = dict(
    X_train=X_train, X_test=X_test, y_test=y_test,
    calibrated_probs=calibrated_probs,
    precision=precision, recall=recall, thr=thr,
    t_low=t_low, t_high=t_high
)

```

Appendix D – Calibration Influence Diagram

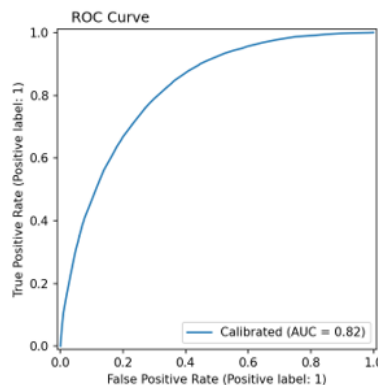


Appendix E --- Influence of calibration on evaluation metrics

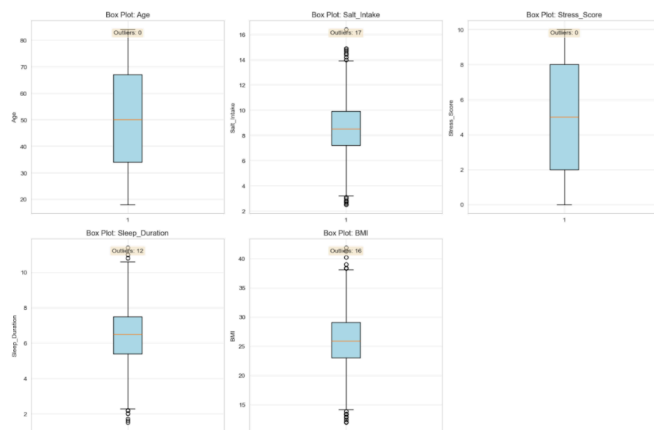
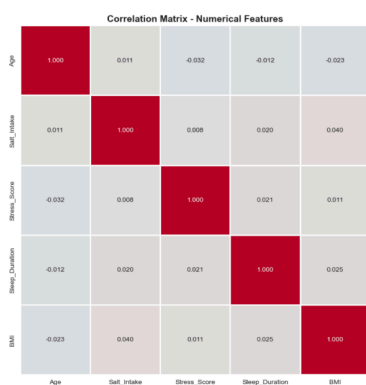
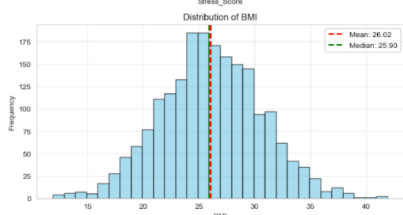
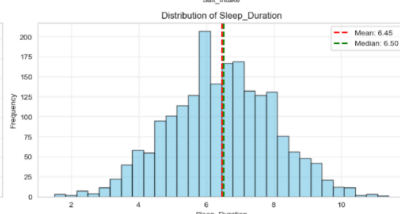
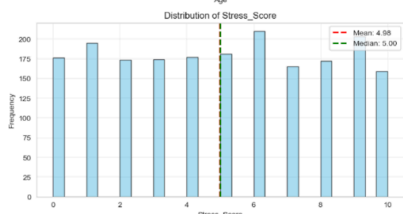
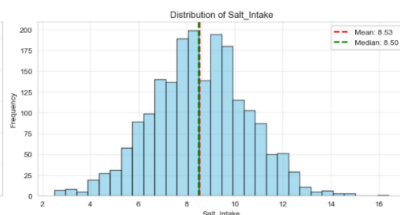
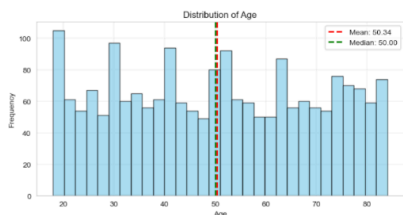
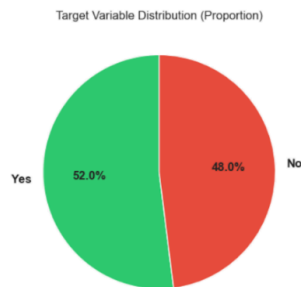
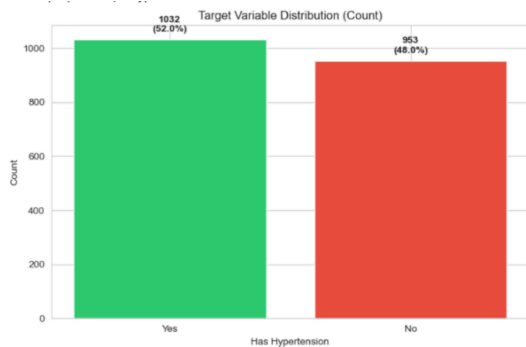
=== BEFORE CALIBRATION ===					
Accuracy: 0.7249					
ROC-AUC: 0.8222					
Brier: 0.1735					
	precision	recall	f1-score	support	
0	0.9441	0.7158	0.8143	42741	
1	0.3373	0.7734	0.4698	7995	
accuracy			0.7249	50736	
macro avg	0.6407	0.7446	0.6420	50736	
weighted avg	0.8485	0.7249	0.7600	50736	

AFTER CALIBRATION					
-Accuracy: 0.8515					
-ROC-AUC: 0.8217					
-Brier: 0.1063					
	precision	recall	f1-score	support	
0	0.8627	0.9796	0.9174	42741	
1	0.6042	0.1665	0.2610	7995	
accuracy			0.8515	50736	
macro avg	0.7334	0.5730	0.5892	50736	
weighted avg	0.8220	0.8515	0.8140	50736	

Appendix F --- ROC Curve of calibrated model



Appendix G -- EDA for hypertension-risk-prediction-dataset



Appendix H --- Evaluation metrics for hypertension model

--- XGBoost Report ---

	precision	recall	f1-score	support
0	0.99	0.96	0.98	191
1	0.97	1.00	0.98	206
accuracy			0.98	397

LogisticRegression Model Report (Test Set):

--- Logistic Regression Report ---

	precision	recall	f1-score	support
0	0.81	0.83	0.82	191
1	0.84	0.82	0.83	206
accuracy			0.82	397

Appendix J --- Example of an use case [1] For diabet patients

```
red_patient = {
    "HighBP": 1,          # Has high blood pressure
    "HighChol": 1,        # Has high cholesterol
    "CholCheck": 1,        # Has checked cholesterol level
    "BMI": 42,             # Body Mass Index indicates obesity
    "Smoker": 1,           # Smokes
    "Stroke": 0,           # No history of stroke
    "HeartDiseaseorAttack": 1, # Has heart disease or history of heart attack
    "PhysActivity": 0,      # Does not engage in physical activity
    "Fruits": 0,           # Does not eat fruits regularly
    "Veggies": 0,          # Does not eat vegetables regularly
    "HvyAlcoholConsump": 0, # Does not consume heavy alcohol
    "AnyHealthcare": 1,     # Has access to healthcare
    "NoDocbcCost": 0,       # Does not avoid doctor due to cost
    "GenHlth": 5,           # General health rated poor (scale 1-5)
    "MentHlth": 20,         # 20 days of poor mental health in past month
    "PhysHlth": 20,         # 20 days of poor physical health in past month
    "DiffWalk": 1,         # Has difficulty walking
    "Sex": 1,              # Male (assuming 1=male, 0=female)
    "Age": 13,             # Age encoded (~65-69 years)
    "Education": 1,        # Low education level
    "Income": 1            # Low income
}
```

```
{'HighBP': 1, 'HighChol': 1, 'CholCheck': 1, 'BMI': 42, 'Smoker': 1, 'Stroke': 0, 'HeartDiseaseorAttack': 1, 'PhysActivity': 0, 'Fruits': 0, 'Veggies': 0, 'HvyAlcoholConsump': 0, 'AnyHealthcare': 1, 'NoDocbcCost': 0, 'GenHlth': 5, 'MentHlth': 20, 'PhysHlth': 20, 'DiffWalk': 1, 'Sex': 1, 'Age': 13, 'Education': 1, 'Income': 1} → Probability=0.645, RiskScore=1.94, Level=RED
```

```
green_patient = {
    "HighBP": 0,          # Normal blood pressure
    "HighChol": 0,        # Normal cholesterol
    "CholCheck": 1,        # Has checked cholesterol level
    "BMI": 28,             # Overweight but not obese
    "Smoker": 0,           # Does not smoke
    "Stroke": 0,           # No history of stroke
    "HeartDiseaseorAttack": 0, # No heart disease
    "PhysActivity": 1,      # Engages in physical activity
    "Fruits": 1,           # Eats fruits regularly
    "Veggies": 1,          # Eats vegetables regularly
    "HvyAlcoholConsump": 0, # Does not consume heavy alcohol
    "AnyHealthcare": 1,     # Has access to healthcare
    "NoDocbcCost": 0,       # Does not avoid doctor due to cost
    "GenHlth": 3,           # General health rated average
    "MentHlth": 2,         # 2 days of poor mental health in past month
    "PhysHlth": 3,         # 3 days of poor physical health in past month
    "DiffWalk": 0,         # Can walk without difficulty
    "Sex": 0,              # Female
    "Age": 8,              # Age encoded (~45-49 years)
    "Education": 5,        # Medium-high education level
    "Income": 6            # Medium income
}
```

```
{'HighBP': 0, 'HighChol': 0, 'CholCheck': 1, 'BMI': 28, 'Smoker': 0, 'Stroke': 0, 'HeartDiseaseorAttack': 0, 'PhysActivity': 1, 'Fruits': 1, 'Veggies': 1, 'HvyAlcoholConsump': 0, 'AnyHealthcare': 1, 'NoDocbcCost': 0, 'GenHlth': 3, 'MentHlth': 2, 'PhysHlth': 3, 'DiffWalk': 0, 'Sex': 0, 'Age': 8, 'Education': 5, 'Income': 6} → Probability=0.101, RiskScore=0.3, Level=GREEN
```

```
low_yellow_patient_A = {
    "HighBP": 1,          # Has high blood pressure
    "HighChol": 0,        # Normal cholesterol
    "CholCheck": 1,        # Has checked cholesterol level
    "BMI": 29,             # Overweight
    "Smoker": 0,           # Does not smoke
    "Stroke": 0,           # No history of stroke
    "HeartDiseaseorAttack": 0, # No heart disease
    "PhysActivity": 1,      # Engages in physical activity
    "Fruits": 1,           # Eats fruits regularly
    "Veggies": 1,          # Eats vegetables regularly
    "HvyAlcoholConsump": 0, # Does not consume heavy alcohol
    "AnyHealthcare": 1,     # Has access to healthcare
    "NoDocbcCost": 0,       # Does not avoid doctor due to cost
    "GenHlth": 3,           # General health rated average
    "MentHlth": 2,         # 2 days of poor mental health
    "PhysHlth": 2,         # 2 days of poor physical health
    "DiffWalk": 0,         # Can walk normally
    "Sex": 0,              # Female
    "Age": 8,              # Age encoded (~45-49 years)
    "Education": 5,        # Medium-high education level
    "Income": 6            # Medium-high income
}
```

```
{'HighBP': 1, 'HighChol': 0, 'CholCheck': 1, 'BMI': 29, 'Smoker': 0, 'Stroke': 0, 'HeartDiseaseorAttack': 0, 'PhysActivity': 1, 'Fruits': 1, 'Veggies': 1, 'HvyAlcoholConsump': 0, 'AnyHealthcare': 1, 'NoDocbcCost': 0, 'GenHlth': 3, 'MentHlth': 2, 'PhysHlth': 2, 'DiffWalk': 0, 'Sex': 0, 'Age': 8, 'Education': 5, 'Income': 6} → Probability=0.182, RiskScore=0.55, Level=YELLOW
```

[2] For hypertension patients

hypertension_test_report		
Actual_Value	Probability	Risk_Level
1	0.2387383135968542	Low Risk (GREEN)
1	0.9998043628696662	High Risk (RED)
0	0.1657264472271723	Low Risk (GREEN)
1	0.9562303764860248	High Risk (RED)
0	0.8274300734823713	High Risk (RED)
0	0.5519438627422354	Medium Risk (YELLOW)
0	0.7312576457655152	High Risk (RED)
0	0.19634410928273902	Low Risk (GREEN)

[3] For stroke patients

EXAMPLES OF PERSONAS

1. High Risk Patient

A 67-year-old male with hypertension and diagnosed heart disease. Married, urban based, employed in the private sector. Blood glucose at 180.5 mg/dL and BMI 32.1 (class I obesity). Former smoker. Displays a strong cluster of risk factors consistent with a high likelihood of cardiovascular and cerebrovascular events.

2. Healthy Young Patient

A 25-year-old male with no hypertension or heart disease. Single, urban resident, private sector worker. Glucose at 90 mg/dL; BMI 22.5. Never smoked. Overall profile reflects excellent health and very low risk.

3. Low Risk Stroke Patient

A 38-year-old female with no hypertension or cardiac issues. Married, self-employed, living in an urban environment. Glucose level: 125 mg/dL; BMI: 24 (normal). Never smoked. Presents a low stroke risk with only mild metabolic contributors.

4. Yellow Patient Low

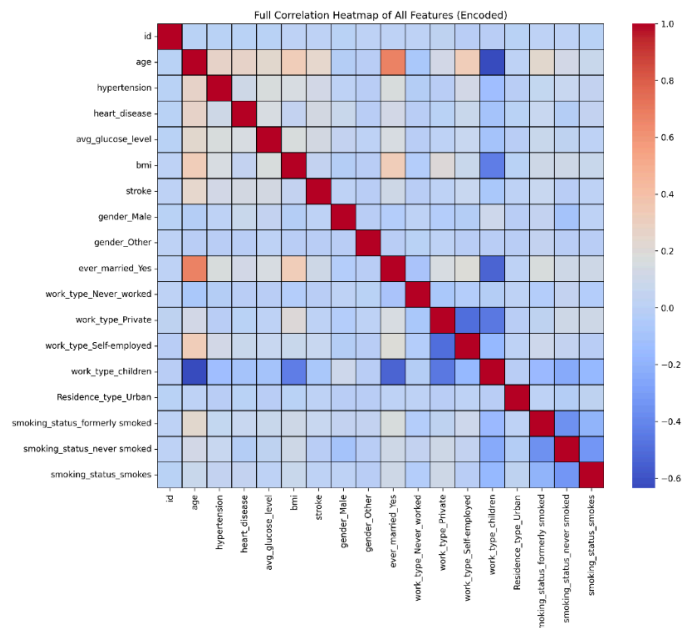
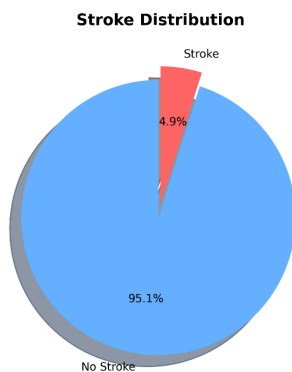
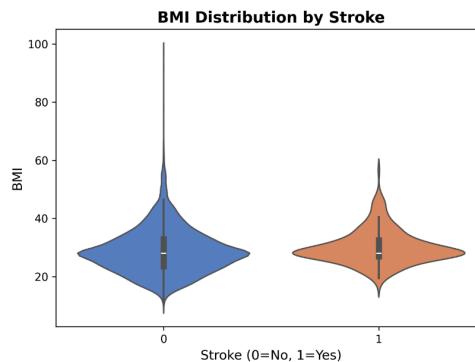
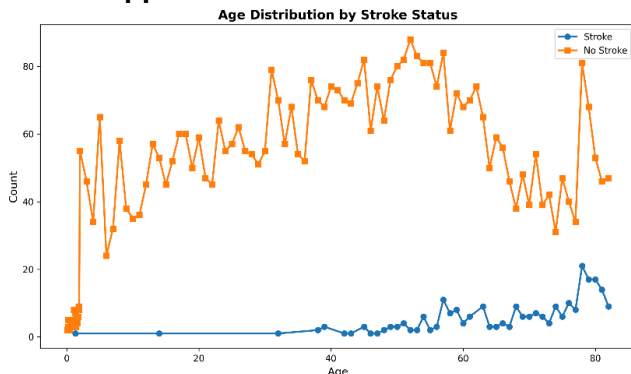
A 45-year-old female. No hypertension or heart disease. Rural resident, self-employed. Blood glucose: 120 mg/dL; BMI: 26 (overweight). Never smoked. Classified as slightly elevated risk.

5. Yellow Patient High

A 55-year-old male with hypertension. Urban resident, employed in the private sector. Glucose: 150 mg/dL; BMI: 30.5 (class I obesity). Former smoker. Presents a moderate to-high risk based on cumulative risk factors.

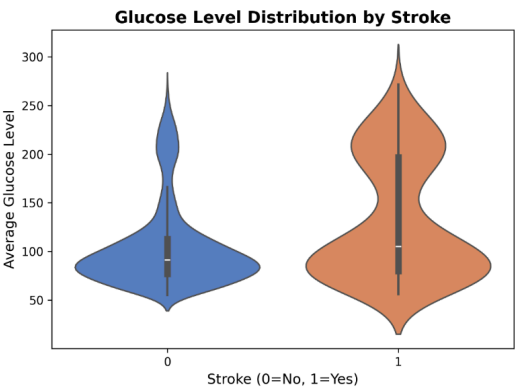
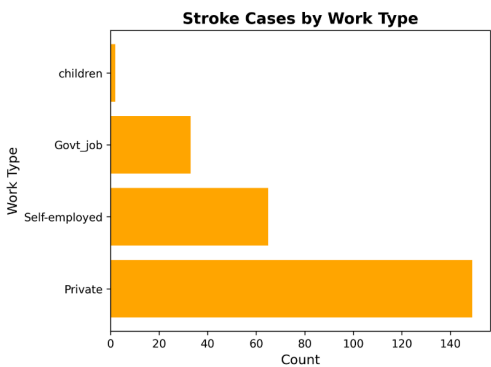
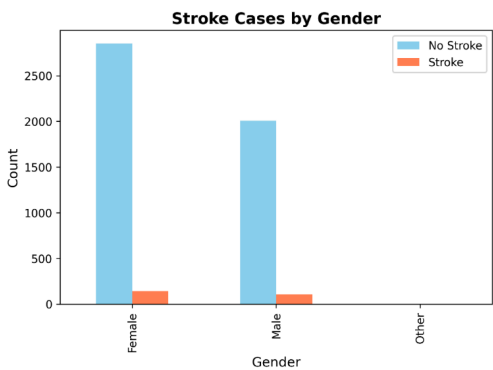
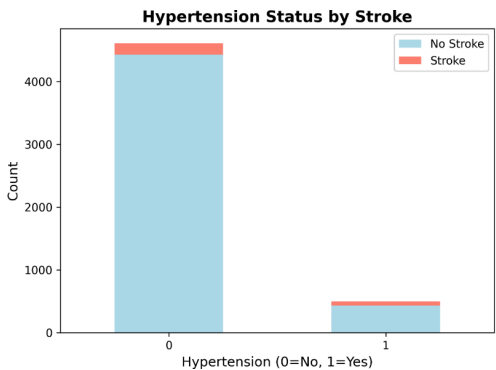
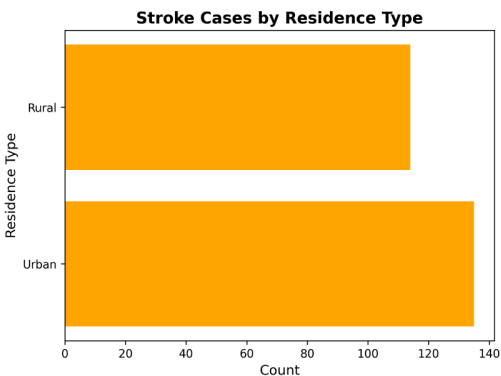
```
Patient high_risk_patient → Probability=0.100, Score=0.3, Level=RED
Patient healthy_young_patient → Probability=0.000, Score=0.0, Level=GREEN
Patient low_risk_stroke_patient → Probability=0.017, Score=0.05, Level=YELLOW
Patient yellow_patient_low → Probability=0.000, Score=0.0, Level=GREEN
Patient yellow_patient_high → Probability=0.094, Score=0.28, Level=YELLOW
```

Appendix I --- EDA for stroke healthcare-dataset-stroke-data.csv dataset



```
Top 5 Positively Correlated Features:
age                0.245257
heart_disease      0.134914
avg_glucose_level  0.131945
hypertension       0.127904
ever_married_Yes   0.108340
Name: stroke, dtype: float64

Top 5 Negatively Correlated Features:
gender_Male        0.009117
smoking_status_smokes 0.008939
id                 0.006388
smoking_status_never smoked -0.004129
gender_Other       -0.003166
Name: stroke, dtype: float64
```



Appendix K --- Evaluation metrics for stroke model

```
AFTER CALIBRATION
-Accuracy: 0.9501
-ROC-AUC: 0.7963
-Brier: 0.0446
```

	precision	recall	f1-score	support
0	0.9510	0.9990	0.9744	972
1	0.0000	0.0000	0.0000	50
accuracy			0.9501	1022
macro avg	0.4755	0.4995	0.4872	1022
weighted avg	0.9045	0.9501	0.9267	1022

	A	B	C	D	E	F
1	feature	impact_stroke_perce	impact_diabetes_pe	impact_hypertensor	distribution	how_effective
2	sudden_onset	80	5	15	0.0027	1
3	paralyzed_area	75	10	15	0.0027	1
4	speech_difficulty	70	15	15	0.0027	1
5	face_numbrness	70	10	20	0.0027	1
6	atital_fibrillation	85	5	10	0.02	1
7	age	60	20	20	1	0.7
8	smoking	55	20	25	0.33	0.7
9	family_history_stroke	70	10	20	0.15	0.68
10	confusion	65	20	15	0.05	0.65
11	frequent_urination	5	90	5	0.15	0.92
12	glucose_lv1	5	90	5	1	0.9
13	excessive_thirst	5	90	5	0.1	0.9
14	family_history_diabe	10	80	10	0.3	0.85
15	bmi	10	70	20	1	0.7
16	frequent_infection	10	80	10	0.2	0.7
17	unintended_weight_	5	85	10	0.05	0.6
18	apptitde	10	75	15	0.1	0.4
19	blood_p	10	10	80	1	1
20	family_history_hyper	15	10	75	0.35	0.88
21	nose_bleed	10	5	85	0.1	0.85
22	cholesterol	30	20	50	0.4	0.7
23	chest_pain	20	10	70	0.2	0.45
24	physical_activity	10	10	80	0.45	0.25
25	headache	20	10	70	0.5	0.3
26	alcohol	25	20	55	0.2	0.2
27	dizziness	25	10	65	0.3	0.2
28	stress	25	15	60	0.8	0.2

```

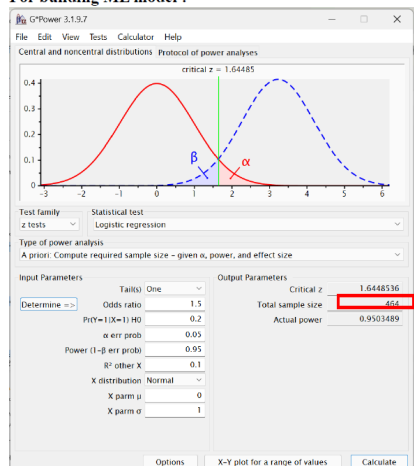
---- Debugging Row 0 ----
unintended_weight_loss | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
paralysed_area         | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
dizziness              | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
glucose_lvl            | val= 0.129 | stroke+= 0.006 | diab+= 0.104 | hyper+= 0.006
sudden_onset           | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
atrial_fibrillation    | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
blood_p                | val= 0.052 | stroke+= 0.005 | diab+= 0.005 | hyper+= 0.042
nose_bleed             | val= 1.000 | stroke+= 0.005 | diab+= 0.043 | hyper+= 0.722
alcohol                | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
frequent_urination     | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
age                    | val= 0.135 | stroke+= 0.057 | diab+= 0.019 | hyper+= 0.019
chest_pain              | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
cholesterol             | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
smoking                 | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
physical_activity       | val= 1.000 | stroke+= 0.025 | diab+= 0.025 | hyper+= 0.200
headache                | val= 1.000 | stroke+= 0.060 | diab+= 0.030 | hyper+= 0.210
excessive_thirst        | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
family_history_diabetes | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
stress                  | val= 1.000 | stroke+= 0.050 | diab+= 0.030 | hyper+= 0.120
family_history_hypertension | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
family_history_stroke   | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
appetite                | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
frequent_infection      | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
face numbness           | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
confusion               | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
speech_difficulty       | val= 0.000 | stroke+= 0.000 | diab+= 0.000 | hyper+= 0.000
bmi                     | val= 0.103 | stroke+= 0.007 | diab+= 0.050 | hyper+= 0.014

--- Final scores for this row ---
Stroke: 0.295, Diabetes: 0.306, Hypertension: 1.333

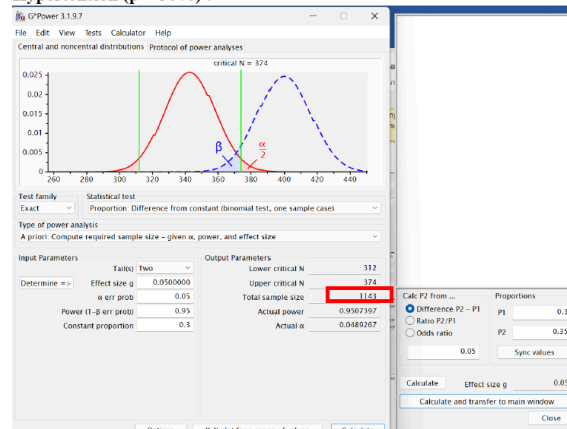
```

Appendix L

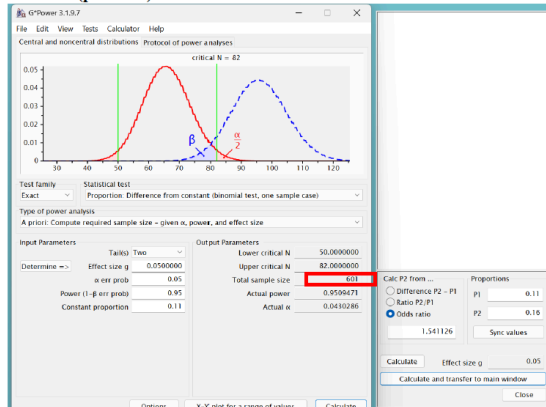
For building ML model :



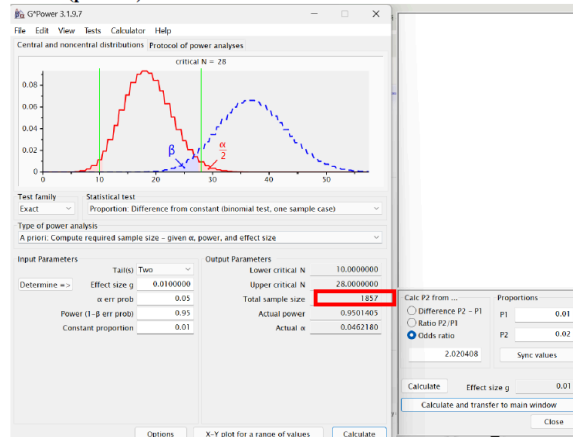
Hypertension (p = 30%) :



Diabetes (p = 16%) :



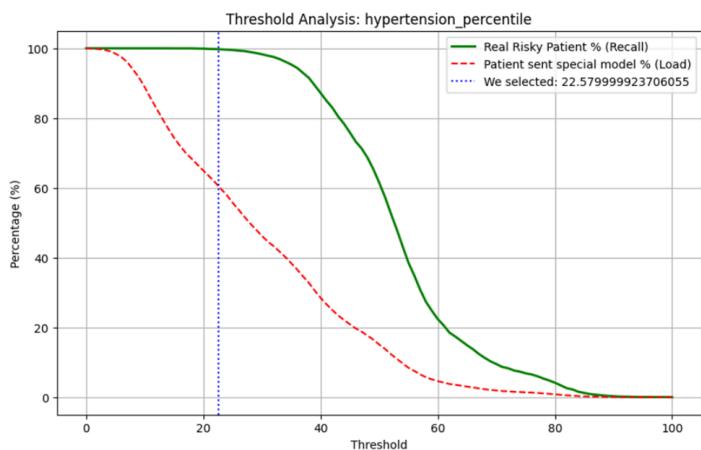
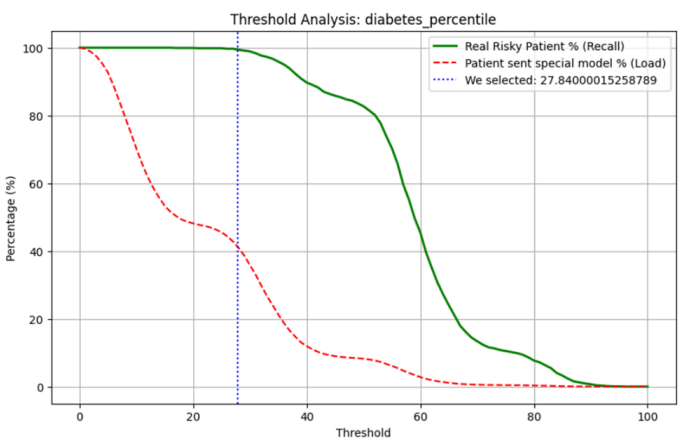
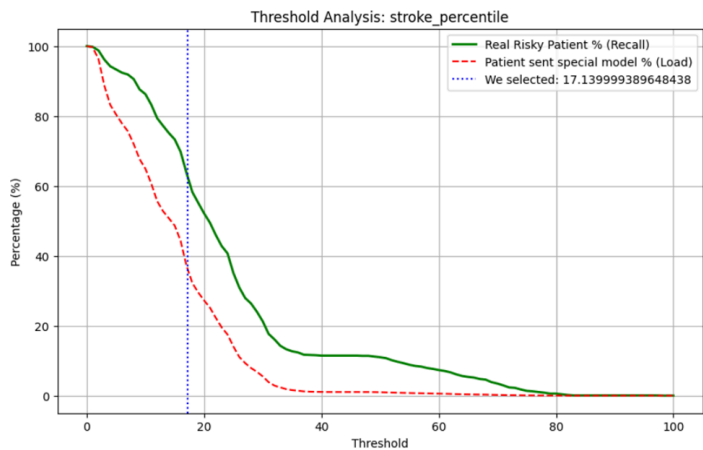
Stroke (p = 1%) :



Appendix M

Dataset Name	Feature Count	Target Labels	Version	Primary Usage
all_data_v1.csv	27	Stroke, Diabetes, Hypertension	v1	Comparison / Legacy
all_data_v2.csv	27	Stroke, Diabetes, Hypertension	v2	System Test Cases
important_data_v1.csv	12 (4+4+4)	Stroke, Diabetes, Hypertension	v1	Comparison
important_data_v2.csv	12 (4+4+4)	Stroke, Diabetes, Hypertension	v2	Main Model Training
stroke_data_v1.csv	9	Stroke	v1	Comparison
stroke_data_v2.csv	9	Stroke	v2	Specialized Model
diabetes_data_v1.csv	8	Diabetes	v1	Comparison
diabetes_data_v2.csv	8	Diabetes	v2	Specialized Model
hypertension_data_v1.csv	10	Hypertension	v1	Comparison
hypertension_data_v2.csv	10	Hypertension	v2	Specialized Model

Appendix N



Appendix P
https://docs.google.com/forms/d/1w74roWUI8u-VZPsKmN_Zn9qT8rGfDZXwc_FqF6JntZI/edit

Appendix O



ANKARA YILDIRIM BEYAZIT ÜNİVERSİTESİ (AYBÜ)
SAĞLIK BİLİMLERİ ETİK KURULU
PROJE ONAY BELGESİ



Ankara Yıldırım Beyazıt Üniversitesi Mühendislik ve Doğa Bilimleri Fakültesi Bilgisayar Mühendisliği Bölümü
Başkanlığı öğrencilerinden ABDULLAH YUSUF ERDEMİR'in, "Yapay Zeka Temelli Sağlık Riski Değerlendirmesi ve
Klinik Kaynak Yönetimi" adlı araştırması değerlendirilmiştir.

- ☒ Etik Açından UYGUN Bulunmuştur.
- ☐ Etik Açından GELİŞTİRİLMESİ GEREKLİDİR Kararı Verilmiştir.
- ☐ Etik Açından RET Kararı Verilmiştir.
- ☐ Klinik Araştırmalar Etik Kuruluna Başvurulması Uygundur.
- ☐ Fen ve Mühendislik Bilimleri Etik Kuruluna Başvurusu Uygundur.

SAĞLIK BİLİMLERİ ETİK KURULU KARARI	
Başvuru Tarihi	9.01.2026
Araştırma Kodu	26
Etik Kurul Toplantı Tarihi ve Karar No	22.01.2026 – 01/26

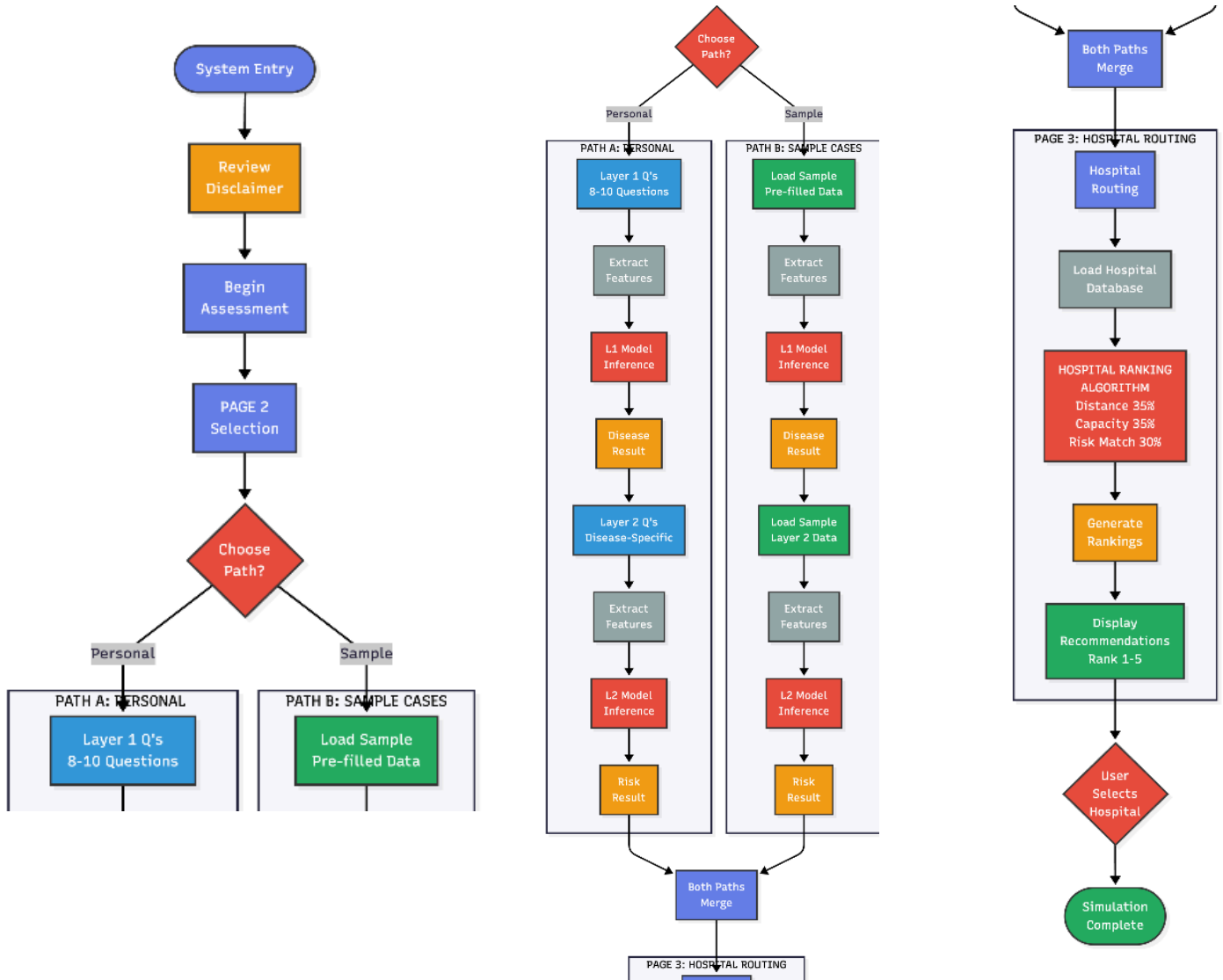
KURUL BAŞKANI, BAŞKAN VEKİLİ VE ÜYELER:

Prof. Dr. BAHAR ANAFOĞLU
Prof. Dr. SEVİL ŞAHİN
Prof. Dr. ÖZDEN YALÇINKAYA ALKAR
Prof. Dr. ŞEYDA TOPRAK ÇELENAZ
Prof. Dr. TAHİR KURTULUŞ YOLDAŞ
Doç. Dr. BERRİN ÇELİK
Doç. Dr. BÜNYAMİN ÇİLDİR
Doç. Dr. ESKAKİLİÇ CEVHAN
Doç. Dr. İZZET İNCE
Doç. Dr. NİMETCAN MEHMET ORHUN
Doç. Dr. NURAL ERZURUM ALİM
Doç. Dr. ŞULE ÇEKİÇ
Dr. Öğr. Üyesi BETÜL AKBUĞA ÖZEL
Dr. Öğr. Üyesi SİTKİ KOCAOĞLU

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Appendix Q



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